Theorem 9: upper bound for the local edit distance

Correspondence between the paper and the Lean 4/Mathlib formalisation

Paper. J. Breitung, T. Gervens, M. Grohe. *Graph Similarity Meets Optimal Transport* (file `Graph_Similarity_Meets_Optimal_Transport.pdf` in the repository). Statement, equation, page and line numbers refer to this PDF; `docs/NUMBERING.md` lists the Lean files and declarations of all numbered statements.

The folder `GsimOT/Theorem9_LocalEditUpperBound` formalises Theorem 9 together with Appendix D.2 (Lemmas 30, 31, Proposition 32 and the path-cutting construction); Theorem 9 is the δ_1 part of Theorem 2. Shared notions and lemmas (Eq. (3), Observation 21, Lemma 29) come from `GsimOT/Common`; they are listed at the end.

Checking. Run `bash build.sh` in the repository root (see `README.md`). The script builds the library and runs `scripts/CheckAxioms.lean`, which must report only the standard axioms `propext`, `Classical.choice`, `Quot.sound`: no `sorry` and no additional axioms.

Main statement in Lean. `GsimOT.theorem9`: assuming Lemma 30 (`CVZRounding`), for all finite graphs G on V and H on W with $|V| = |W|$,

$$\delta_1(G,H) \;\le\; 15\sqrt{\delta_1^{\mathrm{OT}}(G,H)\cdot \ln\bigl(4/\delta_1^{\mathrm{OT}}(G,H)\bigr)}.$$

`GsimOT.theorem9_power`: under the same hypotheses, for every $\alpha < 1/2$ there is C_α with $\delta_1(G,H) \le C_\alpha\,(\delta_1^{\mathrm{OT}}(G,H))^{\alpha}$.

Correspondence: Theorem 9 and its proof

Paper	Page	Statement	Lean declaration(s) and location
Theorem 9	9	$\delta_1(G,H) \le 15\sqrt{\delta_1^{\mathrm{OT}}(G,H)\ln(4/\delta_1^{\mathrm{OT}}(G,H))}$	`GsimOT.theorem9` `Theorem9_LocalEditUpperBound/Theorem9.lean:191`
Theorem 2 (δ_1 part)	3	$\delta_1 \le 15\sqrt{\delta_1^{\mathrm{OT}}\ln(4/\delta_1^{\mathrm{OT}})}$ (Thm. 2 follows from Thms. 6, 9, 11, 14)	`GsimOT.theorem9` `Theorem9_LocalEditUpperBound/Theorem9.lean:191`

after Thm. 9	9	Theorem 9 yields $\delta_1 \le C_\alpha(\delta_1^{\mathrm{OT}})^\alpha$ for every $\alpha < 1/2$	`GsimOT.theorem9_power`
			`Theorem9_LocalEditUpperBound/Theorem9.lean:265`
			`GsimOT.sqrt_mul_log_le_rpow`
			`Theorem9_LocalEditUpperBound/Theorem9.lean:237`

Proof of Thm. 9	30	Same bound with $f = C_1(Q)/n$ for every $Q \in \mathrm{DS}(V,W)$	`GsimOT.theorem9_of_isDS` `Theorem9_LocalEditUpperBound/Theorem9.lean:91`
Appendix D.2, Eq. (18)	28	Reference bijection σ with graph in $\mathrm{supp}(Q)$	`GsimOT.exists_bij_subset_supp` `Common/Birkhoff.lean:20`

Paper	Page	Statement	Lean declaration(s) and location

Appendix D.2, Lemma 29	28	ρ_H and Lemma 29 (shared, see Common)	`GsimOT.rho`
			Common/Lemma29_Integrality.lean:25
			`GsimOT.lemma29_rho_le`
			Common/Lemma29_Integrality.lean:110
			`GsimOT.lemma29_iso`
			Common/Lemma29_Integrality.lean:120
			`GsimOT.lemma29_dOne_eq_zero`
			Common/Lemma29_Integrality.lean:162

Appendix D.2	28	Error $\Phi_v(M) = \sum_{(v',w')\in M} \lvert A_G(v,v') - A_H(\sigma(v),w')\rvert$	`GsimOT.errAt` Theorem9_LocalEditUpperBound/Lemma31_GoodMatching.lean:28

Lemma 30 [CVZ11]	29	Dependent rounding with tail bound (19) — **explicit hypothesis**	`GsimOT.CVZRounding`
			Theorem9_LocalEditUpperBound/Lemma30_DependentRounding.lean:40
			`GsimOT.IsMatching`
			Theorem9_LocalEditUpperBound/Lemma30_DependentRounding.lean:30

Lemma 31	29	Matching $M \subseteq \mathrm{supp}(Q)$ with $n - \lvert M\rvert \le 2\gamma n$, $\Phi_v(M) \le 2L + 10\ln(4n)/\gamma$	`GsimOT.lemma31`
			Theorem9_LocalEditUpperBound/Lemma31_GoodMatching.lean:45
			`GsimOT.IsMatching.card_le`
			Theorem9_LocalEditUpperBound/Lemma31_GoodMatching.lean:33

Proposition 32	29	$d_1 \le 2L + 10\ln(4n)/\gamma + 2\gamma n + n/\ell + 2\ell L$	`GsimOT.proposition32` Theorem9_LocalEditUpperBound/Proposition32_Rounding.lean:153
Proof of Prop. 32	29–30	Rounding step for a given matching (deterministic part)	`GsimOT.prop38_of_matching` Theorem9_LocalEditUpperBound/Proposition32_Rounding.lean:31

Proof of Prop. 32	29	Partial injective map μ; walks $v \to \mu(v)$	`GsimOT.PathCutting.iter`
			Theorem9_LocalEditUpperBound/PathCutting.lean:39
			`GsimOT.PathCutting.PInj`
			Theorem9_LocalEditUpperBound/PathCutting.lean:44
			`GsimOT.PathCutting.NonCycle`
			Theorem9_LocalEditUpperBound/PathCutting.lean:47

Paper	Page	Statement	Lean declaration(s) and location

Proof of Prop. 32	29	Paths are finite and positions along paths are well defined	`GsimOT.PathCutting.lt_card_of_iter`
			Theorem9_LocalEditUpperBound/PathCutting.lean:115
			`GsimOT.PathCutting.depth`
			Theorem9_LocalEditUpperBound/PathCutting.lean:158
			`GsimOT.PathCutting.depth_step`
			Theorem9_LocalEditUpperBound/PathCutting.lean:173

Proof of Prop. 32	29	Cut every path into blocks of ℓ vertices; at most n/ℓ cuts	`GsimOT.PathCutting.cutSet`
			Theorem9_LocalEditUpperBound/PathCutting.lean:212
			`GsimOT.PathCutting.cutMap`
			Theorem9_LocalEditUpperBound/PathCutting.lean:216
			`GsimOT.PathCutting.card_cutSet_mul_le`
			Theorem9_LocalEditUpperBound/PathCutting.lean:237
			`GsimOT.PathCutting.iter_cutMap_eq_none`
			Theorem9_LocalEditUpperBound/PathCutting.lean:291

Proof of Prop. 32	29	Close every block to a cycle; permutation τ	`GsimOT.PathCutting.exists_perm_close` Theorem9_LocalEditUpperBound/PathCutting.lean:344
Proof of Prop. 32	29	τ retains μ outside $\le m + n/\ell$ closing edges; closing edges span $< \ell$ steps	`GsimOT.PathCutting.exists_perm_of_partialInj` Theorem9_LocalEditUpperBound/PathCutting.lean:419
Proof of Prop. 32	29–30	$\rho_H(\sigma(v), \pi(v)) \le 2\ell L$ and $L_{P_\pi}(v, \pi(v)) \le \Phi_v(M) + m + n/\ell + 2\ell L$	`GsimOT.prop38_of_matching` Theorem9_LocalEditUpperBound/Proposition32_Rounding.lean:31
Proof of Thm. 9	30	$L < 1 \Rightarrow G \cong H$ by Lemma 29, i.e. $\delta_1 = 0$	`GsimOT.lemma29_dOne_eq_zero` Common/Lemma29_Integrality.lean:162
Proof of Thm. 9	30	Trivial bound $d_1 \le n$ (case $\gamma^* > 1/2$)	`GsimOT.dOne_le_card` Theorem9_LocalEditUpperBound/Theorem9.lean:34
Proof of Thm. 9	30	$x \mapsto \ln(4x)/x$ is decreasing for $x \ge 1$	`GsimOT.log_four_mul_div_antitone` Theorem9_LocalEditUpperBound/Theorem9.lean:40
Proof of Thm. 9	30	Eq. (20) and normalisation: $b \le 4f + 2\sqrt{2f} + 4\sqrt{5}\sqrt{f \ln(4/f)}$	`GsimOT.theorem9_of_isDS` Theorem9_LocalEditUpperBound/Theorem9.lean:91
Proof of Thm. 9	30	Final numerical estimate $\frac{b}{\sqrt{fh}} \le \frac{4+2\sqrt{2}}{\sqrt{\ln 4}} + 4\sqrt{5} < 15$	`GsimOT.final_estimate` Theorem9_LocalEditUpperBound/Theorem9.lean:57

Shared notions and lemmas used (from `GsimOT/Common`)

Paper	Page	Statement	Lean declaration(s) and location
Section 2	5	Adjacency matrix $A_G \in \{0,1\}^{V\times V}$ (real entries)	`GsimOT.adj` `Common/Basic.lean:41`

Appendix A	14	Degree $\deg_G(v) = \lvert N_G(v)\rvert$; $\sum_{v'} A_G(v,v') = \deg_G(v)$	`GsimOT.deg`
			`Common/Basic.lean:45`
			`GsimOT.sum_adj_eq_deg`
			`Common/Basic.lean:69`

Appendix A, Eq. (6)	14	Discrepancy $D_{G,H}(v,w,v',w') = A_G(v,v') - A_H(w,w')$	`GsimOT.disc` `Common/Basic.lean:78`
Section 2	5	$\mathrm{DS}(V,W)$: doubly stochastic matrices	`GsimOT.IsDS` `Common/Basic.lean:85`

Section 2	5	Permutation matrix $P_\pi \in \mathrm{Perm}(V,W)$; $P_\pi \in \mathrm{DS}(V,W)$	`GsimOT.permMat`
			`Common/Basic.lean:91`
			`GsimOT.permMat_isDS`
			`Common/Basic.lean:93`

Section 2	5	Local cost $L_Q(v,w) = \sum_{(v',w')} \lvert D(v,w,v',w')\rvert\, Q(v',w')$	`GsimOT.localCost` `Common/Basic.lean:131`

Section 2, Eq. (7)	5, 14	$C_1(Q) = \max_{(v,w)\in\mathrm{supp}(Q)} L_Q(v,w)$	`GsimOT.costOne`
			`Common/Basic.lean:137`
			`GsimOT.localCost_le_costOne`
			`Common/Basic.lean:176`
			`GsimOT.costOne_le`
			`Common/Basic.lean:182`

Section 2	5	$C_1(\pi) = C_1(P_\pi)$ and
		$L_{P_\pi}(v,\pi(v)) = \lvert \pi(N_G(v)) \triangle N_H(\pi(v))\rvert$
			`GsimOT.costOneBij`
			`Common/Basic.lean:141`
			`GsimOT.localCost_permMat_apply`
			`Common/Basic.lean:207`
			`GsimOT.costOneBij_le_iff`
			`Common/Basic.lean:212`

Paper	Page	Statement	Lean declaration(s) and location
Section 1, Section 2	1, 5	$d_1(G,H) = \min_{\pi \in \mathrm{Bij}(V,W)} C_1(\pi)$	`GsimOT.dOne` Common/Basic.lean:144
Section 2	5	$d_1^{\mathrm{OT}}(G,H) = \min_{Q \in \mathrm{DS}(V,W)} C_1(Q)$ (as an infimum over $\mathrm{DS}(V,W)$)	`GsimOT.dOTOne` Common/Basic.lean:148
Section 1, Section 2	1, 5	Normalisations $\delta_1 = n^{-1} d_1$, $\delta_1^{\mathrm{OT}} = n^{-1} d_1^{\mathrm{OT}}$ (scaling by $1/n$)	`GsimOT.deltaOne` Common/Basic.lean:151 `GsimOT.deltaOTOne` Common/Basic.lean:154
Section 1, Section 3	3, 6	$d_1^{\mathrm{OT}} \le d_1$ (every bijection is a fractional alignment)	`GsimOT.dOTOne_le_dOne` Common/Basic.lean:264
Section 2, Eq. (3)	5	Birkhoff–von Neumann: $\mathrm{supp}(Q)$ contains the graph of a bijection	`GsimOT.exists_bij_subset_supp` Common/Birkhoff.lean:20
Observation 21	20	$L_Q(v,w) \ge \lvert \deg_G(v) - \deg_H(w) \rvert$	`GsimOT.observation21` Common/Observation21_DegreeBound.lean:21 `GsimOT.observation21_bij` Common/Observation21_DegreeBound.lean:40
Lemma 22 (construction)	21	Adding the clique K ($m = 2\Delta + 2\tau + k(\tau+\delta) + 1$) and the staircase	`GsimOT.lockSize` Common/Lemma22_LockingMarkers.lean:36 `GsimOT.stair` Common/Lemma22_LockingMarkers.lean:40 `GsimOT.lockGraph` Common/Lemma22_LockingMarkers.lean:56
Lemma 22 (1)	21	$C_1(\pi) < \tau \Rightarrow \pi$ fixes all markers and $\pi(V) = V$	`GsimOT.lemma22_bij` Common/Lemma22_LockingMarkers.lean:161
Lemma 22 (2)	21	$Q(z_j, z_j) = 1 \Rightarrow C_1(Q')$ w.r.t. (G',H') equals $C_1(Q)$ w.r.t. (G,H)	`GsimOT.extendId` Common/Lemma22_LockingMarkers.lean:260 `GsimOT.lemma22_cost` Common/Lemma22_LockingMarkers.lean:351

Paper	Page	Statement	Lean declaration(s) and location
Appendix D.2	28	Pseudometric $\rho_H(w,w') = \lvert N_H(w) \triangle N_H(w') \rvert$ and the triangle inequality	`GsimOT.rho` Common/Lemma29_Integrality.lean:25 `GsimOT.rho_triangle` Common/Lemma29_Integrality.lean:36
Lemma 29 (first part)	28	$Q(v,w) > 0 \Rightarrow \rho_H(\sigma(v), w) \le 2L$	`GsimOT.rho_le_localCost_add` Common/Lemma29_Integrality.lean:43 `GsimOT.lemma29_rho_le` Common/Lemma29_Integrality.lean:110
Lemma 29 (second part)	28	$L < 1 \Rightarrow \sigma$ is an isomorphism	`GsimOT.rho_eq_zero_of_deg_eq_of_lt_two` Common/Lemma29_Integrality.lean:79 `GsimOT.lemma29_iso` Common/Lemma29_Integrality.lean:120 `GsimOT.lemma29_dOne_eq_zero` Common/Lemma29_Integrality.lean:162
Lemma 29 ("in particular")	28	$G \ncong H \Rightarrow d_1^{\mathrm{OT}}(G,H) \ge 1$	`GsimOT.lemma29_one_le_dOTOne` Common/Lemma29_Integrality.lean:176

Notes on the formalisation

1. **Lemma 30 is a hypothesis.** Lemma 30 is the dependent rounding theorem of Chekuri, Vondrák and Zenklusen [CVZ11, Thm. 1.1], which the paper cites without proof. It is stated as the proposition `CVZRounding` and appears as the explicit hypothesis `hCVZ` of `theorem9` (no Lean `axiom` is introduced). It is specialised to doubly stochastic Q (a point of the perfect matching polytope), which is the only case used, and a "random matching" is a finitely supported probability distribution on subsets of $V \times W$. Apart from this specialisation, the statement is the paper's (p. 29, l. 895–898).

2. Final estimate of the proof of Theorem 9 (p. 30, l. 945–947): the paper's bound $\frac{b}{\sqrt{fh}} \le 4\sqrt{f/h} + \frac{2\sqrt{2}}{\sqrt{h}} + 4\sqrt{5} \le \frac{4+2\sqrt{2}}{\sqrt{\ln 4}} + 4\sqrt{5} < 15$ (using $f \le 1$ and $h \ge \ln 4$; the constant is ≈ 14.74) is formalised as `final_estimate`.

3. In the proof of Theorem 9, the quantity $\sqrt{n/(2L)}$ is called s in Lean, so $2\sqrt{2nL} = 4Ls$.

4. Optimal vs. arbitrary Q: Lemmas 29, 31 and Proposition 32 are stated for any $Q \in \mathrm{DS}(V,W)$ whose supported local costs are at most L. The paper takes Q optimal and $L = d_1^{\mathrm{OT}}$. Theorem 9 is first proved with $f = C_1(Q)/n$ for every Q (`theorem9_of_isDS`), and the statement for δ_1^{OT} (an infimum in Lean) follows by continuity of $f \mapsto 15\sqrt{f \ln(4/f)}$ at $f > 0$. The case $d_1^{\mathrm{OT}} < 1$ is the isomorphic case (Lemma 29). The paper distinguishes $L = 0$ and $L > 0$ and then uses $L \ge 1$; the Lean proof splits directly into $L < 1$ and $L \ge 1$.

5. Convention $f = 0$: in Lean $\ln 0 = 0$ and $4/0 = 0$, so $15\sqrt{f \ln(4/f)} = 0$ for $f = 0$, matching the paper's convention.

6. Proof of Proposition 32: the path decomposition is formalised in two steps, exactly as described: (i) the outgoing edge of every vertex at position $p \equiv \ell - 1 \pmod{\ell}$ on a path is removed (this cuts every path into consecutive blocks of ℓ vertices plus a final shorter block), and (ii) every block is closed to a cycle. The bound "number of closing edges $\leq m + n/\ell$" is proved by counting the $\leq n/\ell$ cut positions (each owns the ℓ vertices of its block) plus the m path ends. The paper counts $\sum_j \lceil p_j/\ell \rceil \leq m + n/\ell$ instead; both give the same bound.

7. In Proposition 32 the bound $\rho_H(\sigma(v_k), \sigma(v_1)) \leq 2(k-1)L$ is used in the weaker form $\leq 2\ell L$, as in the paper's statement of the claim.

8. Lemma 31 handles the degenerate case $n = 0$ separately (the paper implicitly has $n \geq 1$ since $L > 0$).

9. The remark after Theorem 9 (p. 9, l. 320) that Theorem 9 yields $\delta_1 \leq C_\alpha(\delta_1^{\mathrm{OT}})^\alpha$ for every $\alpha < 1/2$ is `theorem9_power` (with $C_\alpha = 15\sqrt{4^\varepsilon/\varepsilon}$, $\varepsilon = 1 - 2\alpha$), derived from Theorem 9 via $\ln x \leq x^\varepsilon/\varepsilon$ (`sqrt_mul_log_le_rpow`).